
$$0.12 \pm 0.03 0.23 \pm 0.10 \times 0.04 80.053 \times 0.14 70.051 \approx 0.0000.150 \pm 0.019 r = 0.995 \times$$

$$\times n = 5 \times$$

$$\times$$

$$\theta_0 \in \mathbb{R}^d d \theta \in \mathbb{R}^d \ell_2$$

$$\Delta W(\theta,\theta_0)=\frac{\|\theta-\theta_0\|_2}{\|\theta_0\|_2}\times 100$$

$$AB\theta_A\theta_B$$

$$\Delta W(\theta_A,\theta_B)=\frac{\|\theta_A-\theta_B\|_2}{\|\theta_0\|_2}\times 100$$

$$\theta_A,\theta_B\in\mathbb{R}^d \alpha\in[0,1]$$

$$\theta(\alpha) = (1-\alpha)\cdot \theta_A + \alpha\cdot \theta_B$$

$$\mathcal{L}(\theta(\alpha))$$

$$\mathcal{B}(\theta_A,\theta_B)=\mathop{\mathcal{L}(\theta(\alpha))}_{\alpha\in[0,1]}-\frac{\mathcal{L}(\theta_A)+\mathcal{L}(\theta_B)}{2}$$

$$\alpha \in \{0.0,0.1,\ldots,1.0\}$$

$$L\theta^{(\ell)}\subset \theta \ell$$

$$\Delta W^{(\ell)} = \frac{\|\theta^{(\ell)} - \theta_0^{(\ell)}\|_2}{\|\theta_0^{(\ell)}\|_2} \times 100, \quad \ell = 0,1,\ldots,L-1$$

$$r$$

$$r=\frac{\sum_{\ell}(\Delta W_A^{(\ell)}-\overline{\Delta W}_A)(\Delta W_B^{(\ell)}-\overline{\Delta W}_B)}{\sqrt{\sum_{\ell}(\Delta W_A^{(\ell)}-\overline{\Delta W}_A)^2}\sqrt{\sum_{\ell}(\Delta W_B^{(\ell)}-\overline{\Delta W}_B)^2}}$$

$$\theta = \theta_0 + \sigma \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, d)$$

$$\sigma \Delta W(\theta, \theta_0) \mathcal{B}(\theta_0, \theta)$$

$$\mathcal{S} \subseteq \{0,\ldots,L-1\}$$

$$\theta_S^{(\ell)}(\alpha)=\begin{cases} (1-\alpha)\theta_A^{(\ell)}+\alpha\theta_B^{(\ell)} & \ell\in\mathcal{S} \\ \theta_A^{(\ell)} & \ell\notin\mathcal{S} \end{cases}$$

$$\mathcal{B}_{\mathcal{S}} = \mathcal{B}(\theta_{\mathcal{S}}(0), \theta_{\mathcal{S}}(1)) \mathcal{S}$$

$$\theta^{(t)}t=40,80,\ldots,400$$

$$\mathcal{B}(t)=\mathcal{B}(\theta_0,\,\theta^{(t)})$$

$$\Delta W \rightarrow \mathcal{B}$$

$$td$$

$$d=\frac{\bar{x}_1-\bar{x}_2}{s_p},\qquad s_p=\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}$$

$$|d|>0.8|d|>0.5|d|>0.2$$

$$\Delta W \approx 1.4 \Delta W \approx 8$$

$$r=0.995\sim$$

$$\Delta W \approx 1.4 \sim$$

$$\Delta W \approx 8 \times$$

$$\times \times$$

$$\Delta W \approx 5 \Delta W \approx 4.6 \times$$

$$\sim$$

$$\beta_1=0.9,\beta_2=0.999$$

$$\begin{array}{ccccc} & & \pm\sigma & & \\ & \Delta W & \Delta W & \leftrightarrow & \\ 1.4\pm0.0 & 1.5\pm0.1 & 2.0\pm0.1 & & \\ 8.0\pm0.3 & 8.5\pm0.2 & 11.6\pm0.3 & & \end{array}$$

$$\begin{array}{l} \approx \leftrightarrow \leftrightarrow \\ n=5\binom{5}{2}=10 \\ \Delta W \Delta W \approx 8 \end{array}$$

$$\begin{array}{l} \Delta=-0.071\Delta=+0.161 \\ 0.150\pm0.019 \\ \Delta W\rightarrow\Delta W\rightarrow\mathcal{B} \\ \Delta W\approx0.3\Delta W\approx1.1\Delta W\approx1.4 \end{array}$$

$$\begin{array}{l} \Delta W\in\{0.5,1,2,4,8\} \\ \Delta W=8\approx\Delta W=1.5\times \\ r=0.995\times \end{array}$$

$$\begin{array}{l} td \\ p=0.55d=0.49p<0.001d=3.66p<0.001d=4.17p<0.001d=5.27|d|>0.8 \end{array}$$

$$\begin{array}{l} \bar{\boldsymbol{\theta}} = (\boldsymbol{\theta}_A + \boldsymbol{\theta}_B)/2\mathcal{L} \\ \mathcal{B}(\boldsymbol{\theta}_A,\boldsymbol{\theta}_B) \approx \frac{1}{8}\,\Delta\boldsymbol{\theta}^\mathsf{T}\,(\bar{\boldsymbol{\theta}})\,\Delta\boldsymbol{\theta} \end{array}$$

$$\begin{array}{l} \Delta\boldsymbol{\theta}=\boldsymbol{\theta}_B-\boldsymbol{\theta}_A\|\Delta\boldsymbol{\theta}\| \\ \rho=(\Delta\boldsymbol{\theta}^\mathsf{T}\Delta\boldsymbol{\theta}/\|\Delta\boldsymbol{\theta}\|^2)/((\,)/d)\rho=1\rho\gg1\times\Delta W\rho/\rho\approx9 \\ \propto e^{-\delta\ell}(\ell)\propto e^{-\gamma\ell}\mathcal{B}_\ell\propto e^{-(\delta+\gamma)\ell}\delta+\gamma\approx0.080.12 \\ r=0.995 \\ d\boldsymbol{\theta}=-\eta\nabla\mathcal{L}\,dt+\sqrt{\eta}\overline{\boldsymbol{\Sigma}}\,d\boldsymbol{\Sigma}=\|\nabla\mathcal{L}\|^2/\,(\boldsymbol{\Sigma}) \\ \rightarrow\boldsymbol{\Sigma}\rightarrow\rightarrow\rightarrow\rightarrow\boldsymbol{\Sigma}\rightarrow\rightarrow\rightarrow \end{array}$$

$$\pm\sigma$$

$$\begin{array}{cc} 0.053\pm0.011 & 0.051\pm0.013 \\ 0.118\pm0.031 & 0.228\pm0.102 \end{array}$$

$$\begin{array}{c} 0.048\pm0.000 \\ 0.071\pm0.007 \\ 0.087\pm0.032 \\ 0.147\pm0.027\quad\times \end{array}$$

$$P() \approx (-\Delta \boldsymbol{\theta}^\mathsf{T} \Delta \boldsymbol{\theta} / (2\kappa))$$

$$\kappa\kappa\!\rightarrow\!\kappa\!\rightarrow\!\times$$

$$\boldsymbol{\Sigma}$$

$$\lambda>\lambda$$

$$\begin{array}{l} \Delta W \rightarrow \mathcal{B} \\ \Delta W = 8 \Delta W = 1.5 \times \Delta W \end{array}$$

$$+0.061-0.058\times\times$$

$$\begin{array}{l} \rho \\ \times \Delta W \\ r=0.995 \end{array}$$

$$\leftrightarrow$$

$$\Delta W \approx 0.3 \approx 0.001 \Delta W \rightarrow \mathcal{B} \sim$$

$$n=3\sigma^2<0.001n=5n=3n=10\times n=3$$

$$n = 5$$

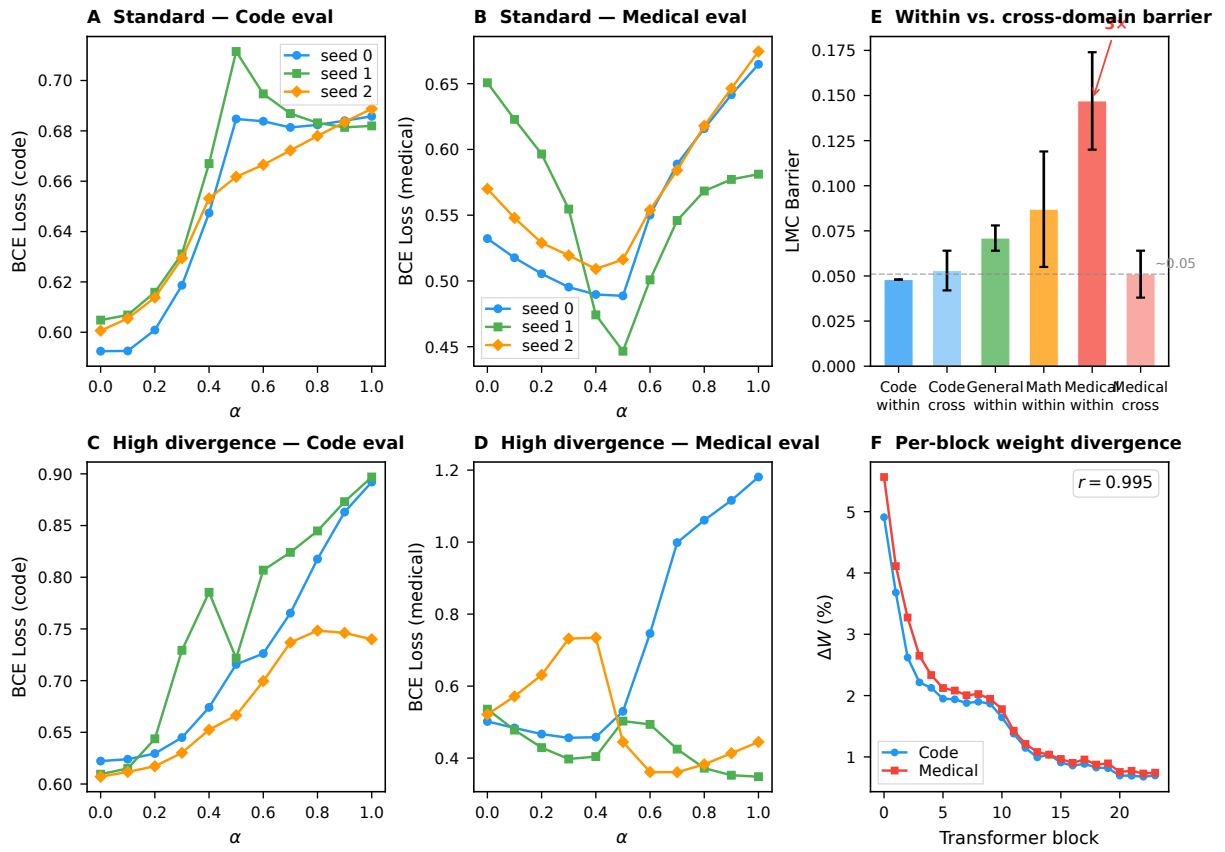
$$\begin{array}{l} \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \leftrightarrow \\ \pm\sigma \quad 0.304 \pm 0.322 \\ \pm\sigma\leftrightarrow \quad 0.203 \pm 0.116 \end{array}$$

$$\begin{array}{ll} 0.599 \pm 0.005 & 0.584 \pm 0.049 \\ 0.686 \pm 0.003 & 0.663 \pm 0.007 \end{array}$$

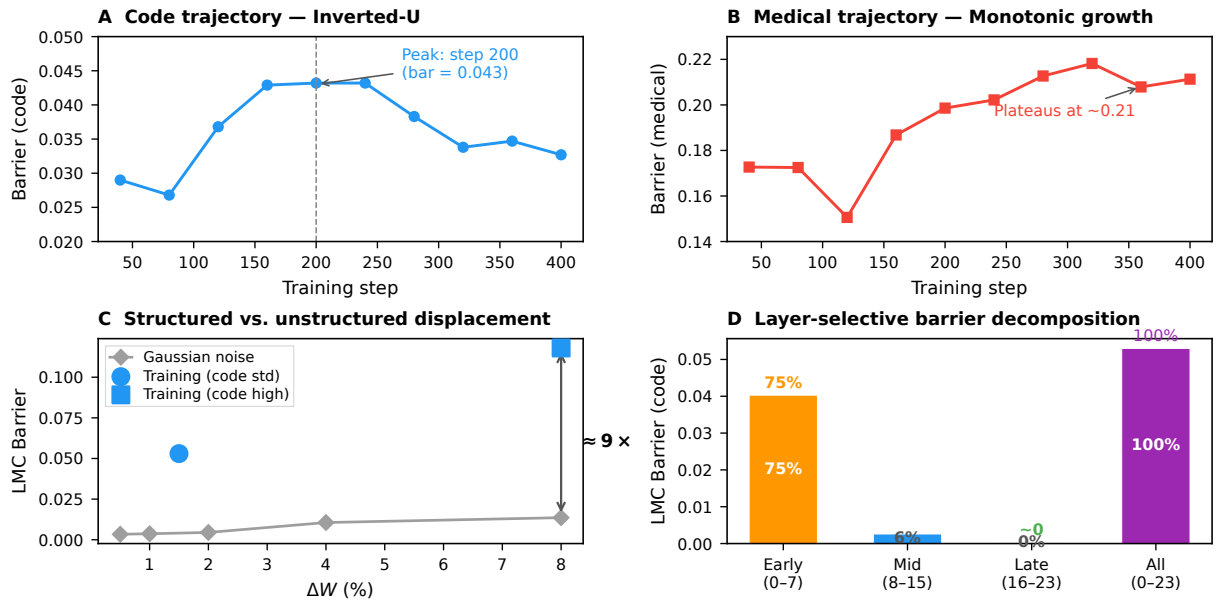
$$\begin{array}{ll} \approx 0.000 & \approx 0.000 \\ \leftrightarrow \quad 0.033 \pm 0.001 & 0.150 \pm 0.019 \end{array}$$

$$\Delta W$$

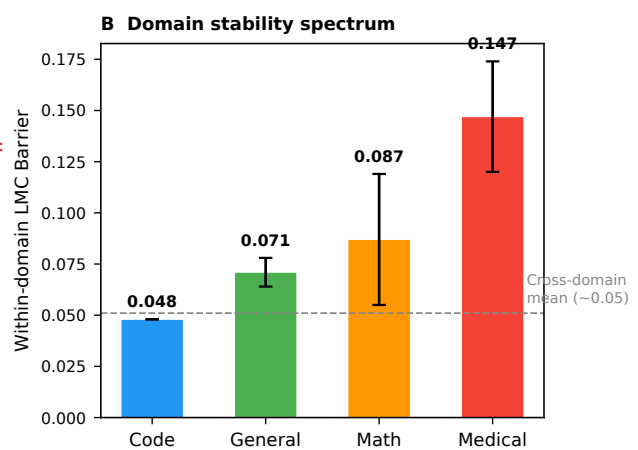
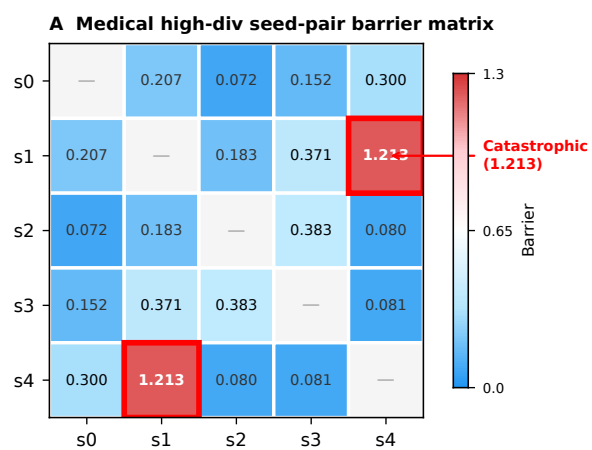
$$\begin{array}{llll} & t & p & d \\ \approx & t(2.0) = 0.60 & & \\ > & t(2.9) = 4.49 & < & \\ \neq & t(2.0) = 5.11 & < & \\ > & t(4.0) = 3.44 & < & \\ & & & \\ > & t(2.5) = 2.83 & & \\ > & t(2.1) = 2.45 & & \end{array}$$



$$\theta(\alpha) = (1 - \alpha)\theta + \alpha\theta \times \pm\sigma r = 0.995$$



$$\Delta W \sim \Delta W \sim \times$$



$$\binom{5}{2} = 10 \times \leftrightarrow \sim$$